Ex.no:7b	# DATA VISUALIZATION USING TABLEAU- GLOBAL TREMOR DATASET

Aim

To connect, analyze, visualize, and create an interactive dashboard using Tableau with the Food Waste Prediction Dataset to analyze food waste patterns, restaurant-wise waste, city-wise waste, food-item distribution, waste percentage, and the factors influencing food waste.

Software Requirements

- Tableau Desktop / Tableau Public
- Microsoft Excel
- CSV Dataset
- Windows Operating System

Theory

Tableau is a powerful Business Intelligence (BI) and Data Visualization tool used to create interactive dashboards and reports.

The Food Waste Prediction Dataset contains information related to food preparation, sales, remaining food, and waste generated by different restaurants.

The dataset includes:

- Date
- Time
- Latitude
- Longitude
- Type
- Depth
- Depth Error
- Depth Seismic Stations
- Magnitude
- Year
- Month
- Month Name
- Magnitude Category
- Magnitude Type
- Magnitude Error
- Magnitude Seismic Stations
- Azimuthal Gap
- Horizontal Distance
- Horizontal Error

Tableau allows users to analyze this data and identify food waste trends, high-waste restaurants, city-wise waste patterns, food-item contributions, and waste percentages.

Tableau Architecture

Global Tremor Dataset
↓
Data Connection
↓
Worksheets
↓
Visualizations
↓
Dashboard
↓
Story

Learning Outcomes

After completing this experiment, students will be able to:

✓ Connect the Global Tremor Dataset to Tableau
✓ Create worksheets using dimensions and measures
✓ Generate different charts and visualizations
✓ Analyze Tremor trends over time
✓ Compare Tremor occurrences across years
✓ Analyze magnitude distribution
✓ Analyze Tremor depth patterns
✓ Analyze geographical Tremor distribution
✓ Apply filters and parameters
✓ Create calculated fields
✓ Build interactive dashboards
✓ Create stories for Tremor analysis

Sample Dataset

Global Tremor Dataset

Date	Latitude	Longitude	Type	Depth	Magnitude	Year	Magnitude Category
1965-01-02	19.246	145.616	Tremor	131.6	6.0	1965	6.0–6.9
1965-01-04	1.863	127.352	Tremor	80.0	5.8	1965	5.5–5.9
1965-01-05	-20.579	-173.972	Tremor	20.0	6.2	1965	6.0–6.9
1965-01-08	-59.076	-23.557	Tremor	15.0	5.8	1965	5.5–5.9
1965-01-10	-13.405	166.629	Tremor	35.0	6.7	1965	6.0–6.9

Algorithm 1 – Connecting Data Source

Step 1

Open Tableau Desktop.

Step 2

Select Connect.

Step 3

Choose Microsoft Excel or Text File depending on the dataset format.

Step 4

Browse and select the Food Waste Prediction Dataset.

Step 5

Open the dataset.

Step 6

Verify the imported fields such as:

- Date
- Time
- Latitude
- Longitude
- Type
- Depth
- Magnitude
- Year
- Month
- Month Name
- Magnitude Category
- Magnitude Type
- Source
- Status

Procedure

Tableau

↓

Connect

↓

Excel / Text File

↓

Global Tremor Dataset

↓

Open

↓

Verify Data

Output

Global Tremor Dataset loaded into Tableau successfully.

Algorithm 2 – Creating Year-wise Tremor Chart

Step 1

Open Sheet 1.

Step 2

Drag **Year → Columns**.

Step 3

Drag **Number of Records → Rows**.

If Number of Records is not available, use:

ID → Rows → Measure → Count

Step 4

Select **Bar Chart** from Show Me.

Step 5

Enable Show Mark Labels.

Step 6

Sort the years based on the number of recorded events if required.

Step 7

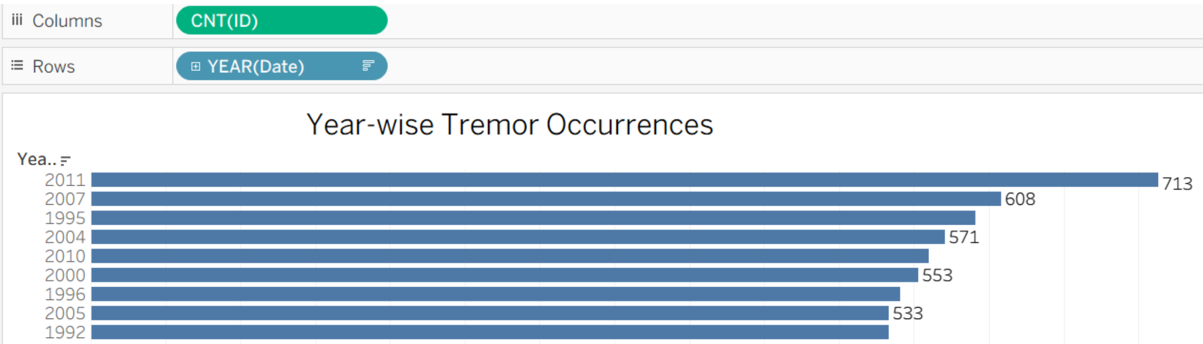
Add the title: **Year-wise Tremor Occurrences**

Output

A bar chart displaying the number of Tremor events recorded in each year.

Interpretation

The bar chart displays the number of seismic events recorded in each year. It helps identify years with relatively higher and lower numbers of recorded events.



Algorithm 3 – Bar Chart Creation

Objective

To analyze the distribution of Tremors according to their magnitude categories.

Step 1

Drag **Magnitude Category → Rows**.

Step 2

Drag **Number of Records → Columns**.

Step 3

Select **Bar Chart** from Show Me.

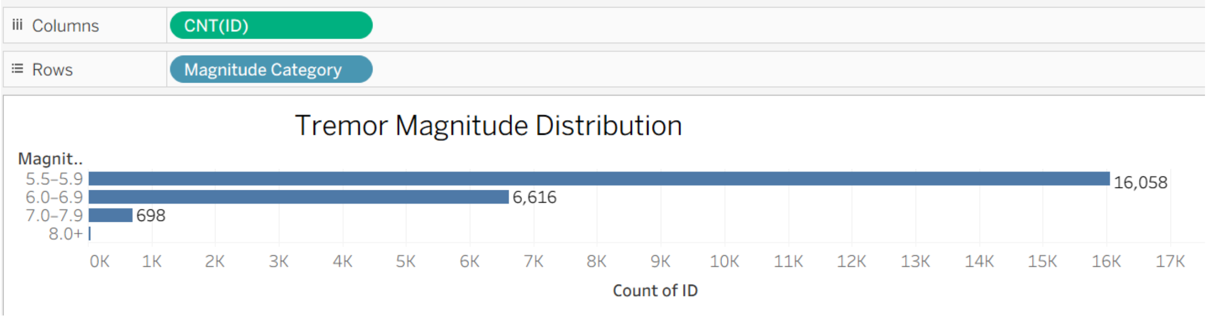
Step 4

Enable Show Mark Labels.

Step 5

Sort the magnitude categories appropriately.

Output Visualization



Interpretation

The bar chart helps identify the most frequently occurring Tremor magnitude category and compare the occurrence of moderate and high-magnitude events.

Algorithm 4 – Pie Chart Creation

Objective

To analyze the distribution of seismic events based on event type.

Step 1

Drag **Type** → **Color**.

Step 2

Select **Pie Chart** from Show Me.

Step 3

Drag **Number of Records** → **Angle**.

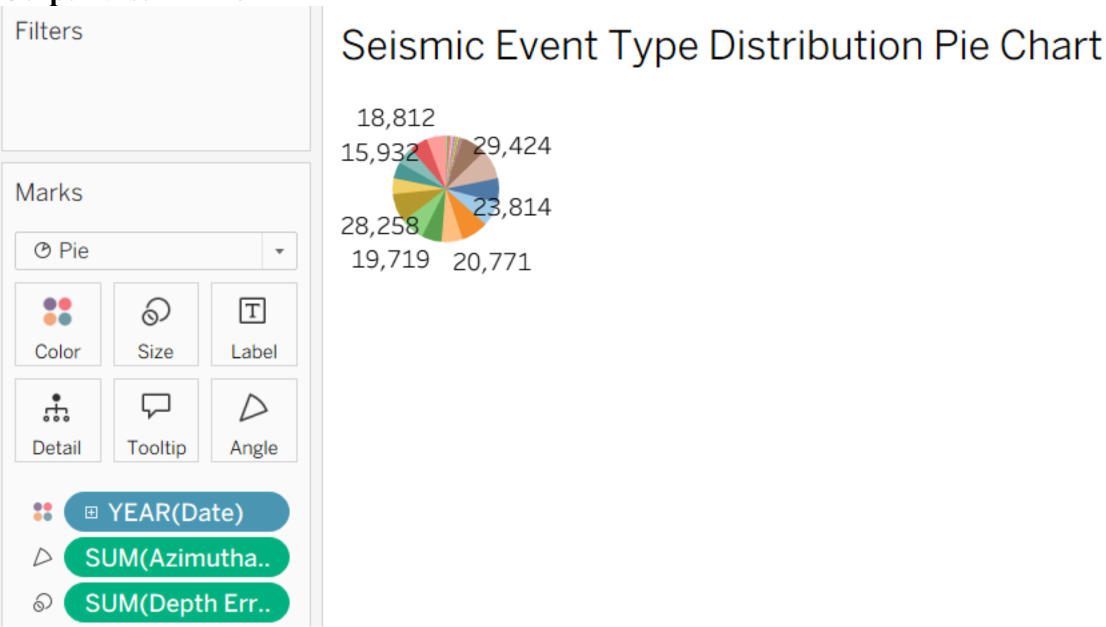
Step 4

Drag **Type** → **Label**.

Step 5

Enable Show Labels.

Output Visualization



Interpretation

The pie chart shows the contribution of different event types to the overall dataset.

Algorithm 5 – Line Chart Creation

Objective

To analyze the trend of Tremor occurrences over time.

Step 1

Drag **Year** → **Columns**.

Step 2

Drag **Number of Records** → **Rows**.

Step 3

Select **Line Chart**.

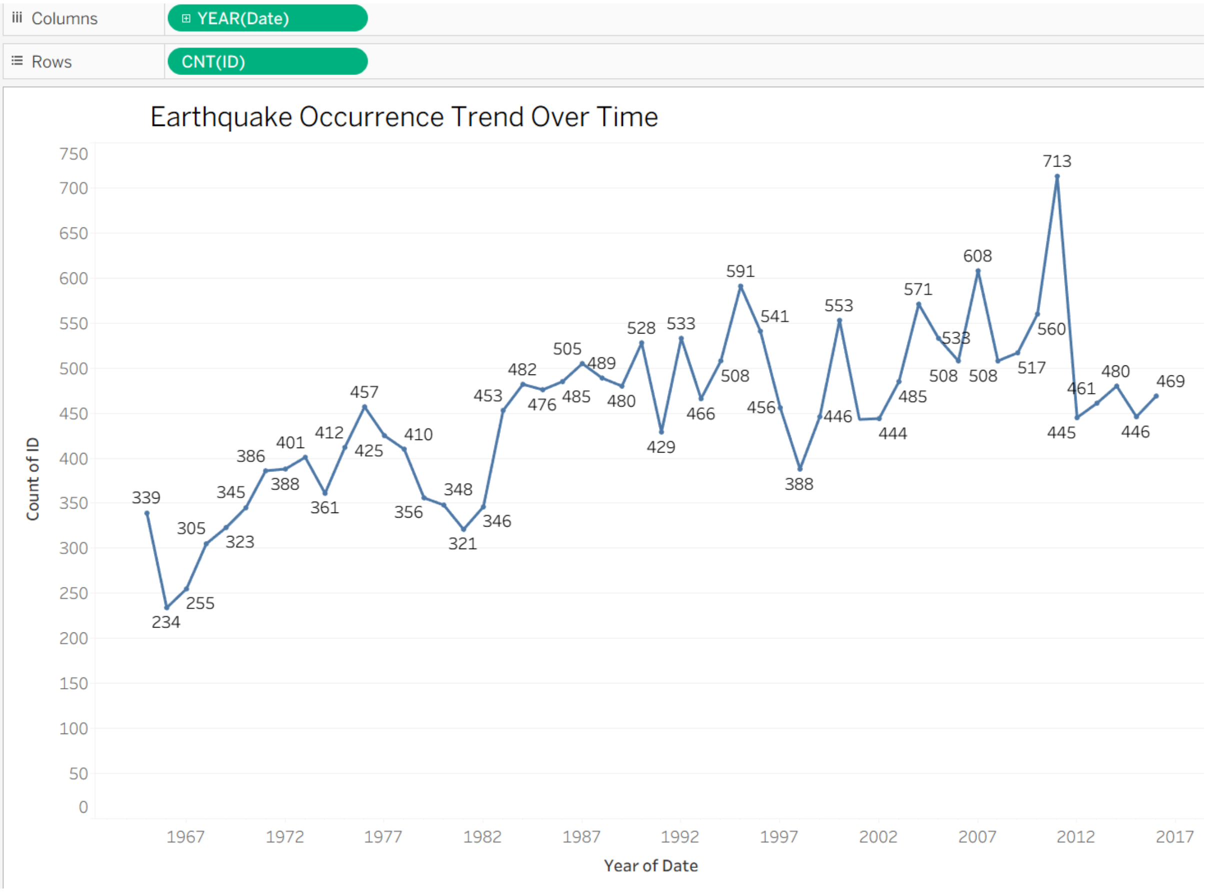
Step 4

Enable data labels if required.

Step 5

Analyze the Tremor occurrence trend over different years.

Output Visualization



Interpretation

The line chart shows how the number of recorded seismic events changes over time.

It helps identify:

- Increasing Tremor occurrence periods
- Decreasing occurrence periods
- Peak years
- Low occurrence years
- Long-term changes in recorded Tremor activity

Algorithm 6 – Creating Calculated Field

Objective

To classify Tremors as shallow, intermediate, or deep based on their depth.

Step 1

Select:

Analysis → Create Calculated Field

Step 2

Enter the field name:

Depth Category

Step 3

Enter the formula:

IF [Depth] < 70 THEN "Shallow"

ELSEIF [Depth] < 300 THEN "Intermediate"

ELSE "Deep"

END**Step 4**

Click OK**.**

Output

Depth	**Depth Category**
20	Shallow
80	Intermediate
131.6	Intermediate
565	Deep

Interpretation

The calculated field helps classify seismic events according to their depth and makes it easier to compare shallow, intermediate, and deep events.

Algorithm 7 – Applying Filters

Objective

To analyze Tremor events for a specific year, magnitude category, or event type.

Step 1

Drag **Year → Filters** shelf.

Step 2

Select the required year.

Step 3

Click Show Filter.

Step 4

Select a particular year such as **2011**.

Step 5

Observe the updated charts.

Example

Filter: Year = 2011

Output

The visualizations display only Tremor records belonging to the selected year.

Interpretation

The Year filter allows users to focus on a particular period and perform detailed Tremor analysis.

Algorithm 8 – Creating Parameter

Objective

To dynamically display Tremors based on a user-defined minimum magnitude value.

Step 1

Select:

Create Parameter

Step 2

Specify the parameter name:

Minimum Magnitude

Step 3

Set the data type as:

Float

Step 4

Set the required minimum value.

Step 5

Create a calculated field using the parameter.

Field name:

Magnitude Threshold Filter

Formula

[Magnitude] >= [Minimum Magnitude]

Output

The visualization displays Tremor records having a magnitude of **7.0 or greater**.

Interpretation

The parameter allows users to dynamically identify high-magnitude Tremor events based on a selected threshold.

Algorithm 9 – Dashboard Creation

Objective

To create an interactive **Global Tremor Analysis Dashboard**.

Step 1

Create multiple worksheets.

Step 2

Select Dashboard → New Dashboard.

Step 3

Drag the required worksheets into the dashboard.

Step 4

Arrange the charts clearly.

Step 5

Add filters for:

- Year
- Month
- Type
- Magnitude Category
- Magnitude Type

- Source
- Status

Step 6

Add the Minimum Waste Amount parameter.

Step 7

Add appropriate titles and labels.

Dashboard Analysis

Total Tremor Events

COUNT(ID)

Shows the total number of records/events.

Average Magnitude

AVG(Magnitude)

Shows the average magnitude of recorded events.

Maximum Magnitude

MAX(Magnitude)

Shows the highest magnitude event in the dataset.

Your dataset's maximum magnitude is 9.1.

Average Depth

AVG(Depth)

Shows the average depth of recorded events.

Tremor Types

Displays the distribution of different event types.

Magnitude Categories

Displays the number of events in each magnitude category.

Interpretation

The dashboard provides a consolidated view of Tremor activity, magnitude distribution, depth patterns, event types, and temporal trends. It allows users to interactively analyze the dataset using filters and parameters.

Algorithm 10 – Story Creation

Objective

To present the analysis of global Tremor activity through a sequence of visual insights.

Step 1

Create the required worksheets.

Step 2

Create the **Global Tremor Analysis Dashboard**.

Step 3

Select Story → New Story.

Step 4

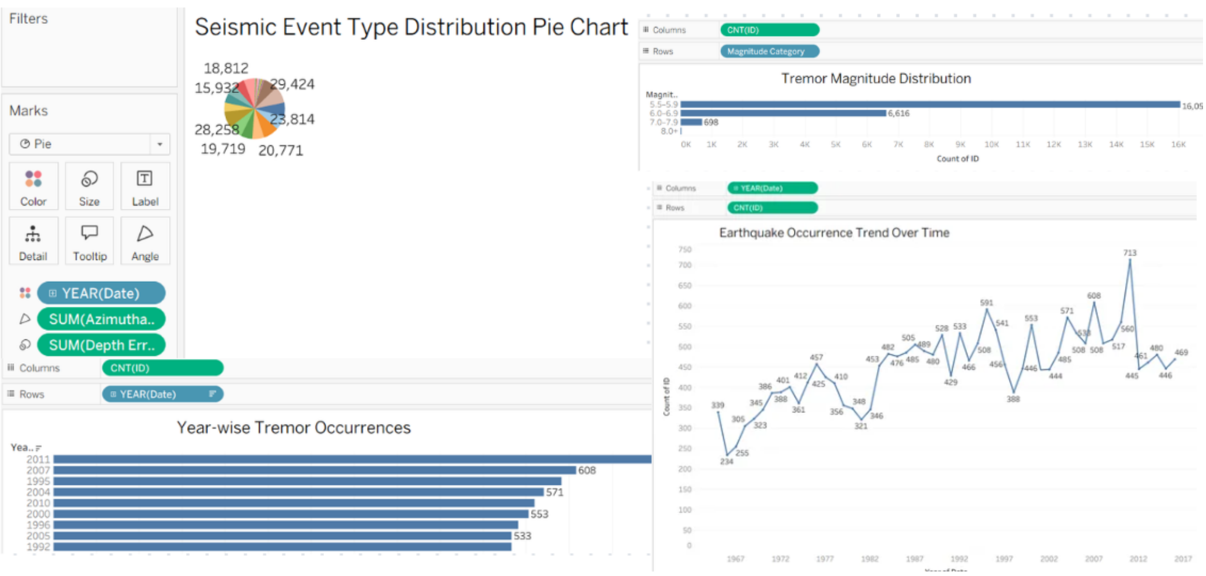
Add the dashboard and required worksheets as story points.

Step 5

Arrange the story points in a logical sequence.

Step 6

Present the findings.



Example Story

Story Point 1

Overall Tremor Analysis

Displays total Tremor events, average magnitude, maximum magnitude, average depth, and magnitude distribution.

Story Point 2

Tremor Trend Over Time

Shows how the number of recorded Tremor events changes across different years.

Story Point 3

Magnitude-wise Tremor Analysis

Shows the distribution of Tremors across different magnitude categories.

Story Point 4

Depth-wise Tremor Analysis

Shows the distribution of shallow, intermediate, and deep Tremors

Story Point 5

Geographical Tremor Analysis

Shows the geographical distribution of seismic events using latitude and longitude.

Story Point 6

Final Tremor Analysis

Summarizes the major patterns and observations obtained from the dataset.

Applications

Tremor Monitoring

Analysis of Tremor occurrences, magnitude, depth, and geographical locations.

Disaster Management

Identification and analysis of high-magnitude Tremor events for improved disaster preparedness.

Seismic Activity Analysis

Analysis of Tremor frequency and magnitude patterns across different years.

Geographical Analysis

Visualization of Tremor locations using latitude and longitude.

Risk Assessment

Identification of high-magnitude and potentially significant seismic events.

Business Intelligence

Interactive visualization and dashboard-based Tremor data analysis.

Decision Making

Identification of significant Tremor patterns for planning and analysis.

Data Analytics

Interactive visualization and exploratory analysis of global seismic activity.

Result